

Abdullah Nayem Wasi Emran

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RESEARCH INTEREST

AI/ML for Healthcare, Computational Biology & Bioinformatics, Causal & Fair ML, Health & Medical Informatics

EDUCATION

- **Bangladesh University of Engineering and Technology (BUET)** Feb 2020 - Mar 2025
BSc. (Engg.) in Computer Science and Engineering Dhaka, Bangladesh
 - **Honors:** Dean's list award and university merit scholarship recipient in four out of eight semesters
 - **Relevant Coursework:** Machine Learning, Computer Security, Introduction to Bioinformatics, Computer Networks, Data Structures and Algorithms, Database, Linear Algebra, Software Engineering

TEACHING EXPERIENCE

- **BRAC University** June 2025 - Present
Lecturer, Department of Computer Science and Engineering Dhaka, Bangladesh
 - **Teaching:** Discrete Mathematics, Compiler Design, Computer Networks

PUBLICATIONS

- **Abdullah Nayem Wasi Emran**, A. B. M. Alim Al Islam, "Domain Adaptive Uplift Modeling across Heterogeneous Mental Health Cohorts," *iScience*, 2026. DOI: [https://www.cell.com/iscience/fulltext/S2589-0042\(26\)01402-1](https://www.cell.com/iscience/fulltext/S2589-0042(26)01402-1).
- **Abdullah Nayem Wasi Emran**, Tanveer Rahman, "SCOPES: Stability-Aware Cross-Platform Feature Selection for Matched TCGA Gene Expression and RNA-Seq Data," AAAI 2026 Workshop on AI to Accelerate Science and Engineering (AI2ASE), 2026. Camera-ready PDF.
- **Abdullah Nayem Wasi Emran**, A. B. M. Alim Al Islam, "Mental Health Across Contexts: A Cross-Dataset Study Covering Medical Students, Quarantined Individuals, and Psychiatric-Disordered Subjects," *Humanities & Social Sciences Communications*, 2025. DOI: 10.1057/s41599-025-05053-x.

MANUSCRIPTS UNDER REVIEW OR IN PREPARATION

- **Abdullah Nayem Wasi Emran**, Majisha Jahan Disha, Rezwana Reaz, "Fusing Exploration and Exploitation: Hybrid Fireworks-Whale Optimization for Multi-Objective Independent Task Scheduling in Cloud Environments," *under review at Cluster Computing (Springer)*, 2025. Preprint: 10.21203/rs.3.rs-6807131/v1

RESEARCH EXPERIENCE

- **Efficient RNA-Protein Interaction Prediction with Foundation-Model Embeddings** July 2025 – Present
Supervised by [Dr. Mohammad Saifur Rahman](#) (Bioinformatics, Deep Learning)
 - Developing RNA-protein interaction prediction models using fixed foundation-model embeddings for RNAs and proteins, with experiments on standard RPI benchmarks and external cold-start settings.
 - Implemented lightweight embedding-based and graph-based architectures, including PairMLP variants and prototype-conditioned graph models, under edge-level, RNA-cold, and protein-cold evaluation protocols.

• Domain-Adaptive Uplift Modeling across Heterogeneous Mental-Health Cohorts

Apr 2025 – Jan 2026

Supervised by [Dr. A. B. M. Alim Al Islam](#)
(Causal Inference, Machine Learning)

- Harmonizing heterogeneous cohort data (students, quarantined, clinical) into a comparable schema.
- Training a domain-adversarial uplift model with fairness-aware regularization and evaluating with leave-one-cohort-out AUUC.
- Studying cohort-specific calibration to handle inverted or unstable treatment effects.

• Hybrid Fireworks–Whale Optimization for Multi-Objective Cloud Task Scheduling

Nov 2023 – Mar 2025

Supervised by [Dr. Rezwana Reaz](#)
(Cloud Computing, Optimization)

- Designing hybrid meta-heuristics that fuse exploration (FWA) with exploitation (WOA) for cloud task scheduling.
- Benchmarking methods and analyzing trade-offs in makespan, monetary cost, and CPU/RAM/bandwidth usage.

• Mental Health Across Contexts: A Cross-Dataset Study

Jul 2024 – Jan 2025

Supervised by [Dr. A. B. M. Alim Al Islam](#)
(Data Mining, Network Analysis)

- Analyzing mental-health outcomes across medical students, quarantined individuals, and psychiatric-patient cohorts.
- Applying statistical modeling, network analysis, and causal methods to uncover robust cross-dataset trends.

ACADEMIC PROJECTS

• SCOPES: Cross-Platform Gene Signature Selection

Bioinformatics Course Project [\[repo\]](#)

- Built a leak-free, multi-objective pipeline (AUC, Kuncheva stability, MMD) to select platform-invariant gene subsets on Agilent that transfer to RNA-Seq without retraining.
- On TCGA-BRCA (505 tumor / 25 normal), achieved $AUC_{\text{Agilent}}=0.69$, $AUC_{\text{RNA-Seq}}=0.61$ ($\Delta AUC=-0.08$) with a 1-gene model; contrasted with a 30-gene model (near-perfect source AUC, poor transfer).
- **Tools & Technology:** Python (scikit-learn, pymoo), NSGA-II, MAD filter, AUC/MMD/Kuncheva

• Lightweight Deep Learning Models for Histopathological Cancer Cell Classification

Machine Learning Course Project [\[repo\]](#)

- Designed and implemented lightweight models using EfficientNet-B0 to classify histopathological cancer cells.
- Integrated attention mechanisms like SEBlock, MSFF, and CBAM to enhance feature recalibration and spatial learning.
- Used transfer learning (PathMNIST) and augmentations, achieving 98% accuracy on CRC-VAL-HE-7K dataset.
- **Tools & Technology:** Python (PyTorch), SEBlock, CBAM, EfficientNet-B0, Matplotlib

• Ray-Tracing Pipeline

Computer Graphics Sessional [\[repo\]](#)

- Developed a raster-based graphics pipeline and implemented ray tracing using C++ and OpenGL.
- Applied Phong Lighting model (ambient, diffuse, specular, recursive reflection) to render realistic shapes.
- **Tools & Technology:** C++, OpenGL

• RecipeShare

Software Development Project [\[repo\]](#)

- Developed a recipe-sharing web application with features for upload, editing, blog posts, and meal planning.
- Implemented recipe generation from images of ingredients using backend processing.
- **Tools & Technology:** Next.js, Django, PostgreSQL

• Building a Subset C Compiler

Compiler Sessional [\[repo\]](#)

- Developed a compiler from scratch in C++ using Bison and Flex.
- Implemented symbol table, lexical analyzer, semantic analyzer, and three-address code generation.
- **Tools & Technology:** C++, Bison, Flex

• BookExchange

ISD Course Project [\[repo\]](#)

- Built a platform for users to sell, borrow, rent, or exchange books with reviews and search functionality.
- **Tools & Technology:** Next.js, Django, PostgreSQL

- **Network Simulations & TCP Adaptive Reno**

Networking Sessional [\[repo\]](#)

- Implemented socket programming in Java for inter-system communication.
- Simulated network architectures with NS-3 and explored TCP Adaptive Reno congestion control.
- **Tools & Technology:** Java, NS-3

AWARDS AND HONORS

- **Dean's List Award and University Merit Scholarship:** Academic excellence at BUET 2022,2023

SKILLS

- **Programming Languages:** Python, C/C++, Java, JavaScript, Bash, MySQL, HTML/CSS
- **Tools & Frameworks:** NS-3, Next.js, Django, Oracle DBMS, Git, LaTeX, Autopsy
- **Libraries:** OpenGL, Scikit-learn, Pandas, Matplotlib, Seaborn
- **Software:** Navicat, Wireshark, MS Office, Adobe Photoshop

REFERENCES

- **Dr. Mohammad Saifur Rahman** Email: mrahman@cse.buet.ac.bd
Professor, Department of CSE, BUET [Graduate Thesis Supervisor]
- **Dr. A. B. M. Alim Al Islam** Email: alim_razi@cse.buet.ac.bd
Professor, Department of CSE, BUET [Research Supervisor]
- **Dr. Rezwana Reaz** Email: rimpi@cse.buet.ac.bd
Associate Professor, Department of CSE, BUET [Undergraduate Thesis Supervisor]